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One-step targeted maximum likelihood estimation for targeting cause-specific absolute risks and survival curves

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Summary

This paper considers the one-step targeted maximum likelihood estimation methodology for multi-dimensional causal parameters in general survival and competing risk settings where event times take place on the positive real line and are subject to right censoring. We focus on effects of baseline treatment decisions possibly confounded by pretreatment covariates, but remark that our work generalizes to settings with time-varying treatment regimes and time-dependent confounding. We point out two overall contributions of our work. First, our methods can be used to obtain simultaneous inference for treatment effects on multiple absolute risks in competing risk settings. Second, our methods can be used to achieve inference for the full survival curve, or a full absolute risk curve, across time. The one-step targeted maximum likelihood procedure is based on a one-dimensional universal least favourable submodel for each cause-specific hazard that we implement in recursive steps along a corresponding nonuniversal multivariate least favourable submodel. Our empirical study demonstrates the practical use of the methods.

Some key words: Average treatment effect; Competing risks; Semiparametric efficiency; Survival analysis; Targeted maximum likelihood estimation.

1. Introduction

In survival and competing risk analysis, semiparametric efficient and doubly robust estimation (Bickel et al., 1993; van der Laan & Robins, 2003; Tsiatis, 2007) provides a promising methodology for drawing inference about the causal effect of treatments (Robins, 1986; Hernan & Robins, 2020). These methods are increasingly used to target causal parameters in place of hazard ratios obtained in a Cox regression analysis (Cox, 1972) that generally fail to capture time-varying effects as well as to reflect a causal effect (Hernán, 2010; Martinussen et al., 2018). Likewise, they provide an alternative to unadjusted Kaplan–Meier estimation (Kaplan & Meier, 1958) of survival curves that yields biased estimation under nonrandomized treatment or covariate-dependent censoring, and may, furthermore, be inefficient in the presence of predictive covariates (Lu & Tsiatis, 2008; Rubin & van der Laan, 2008; Moore & van der Laan, 2009b; Rotnitzky et al., 2012; Díaz et al., 2019).

Often causal estimation focuses on univariate parameters, defined, for example, as the average treatment effect on the survival or an absolute risk probability evaluated at a single predefined point in time (e.g., Andersen et al., 2017; Ozenne et al., 2020). In time-to-event settings, however, such univariate parameters may not always tell the full story. Importantly, when restricting the analysis to a single time-point, one may inevitably fail to discover important time-varying effects. For example, in a reanalysis done by Hernán et al. (2008) it was revealed that conflicting conclusions regarding the harmful or beneficial effect of hormone therapy on coronary heart disease risk in randomized trial and observational data were due to intersecting survival curves and a flawed analysis of the observational data that overlooked an early harmful and later beneficial effect. With such occurrences, the choice of time horizon at which effects are evaluated necessarily makes a critical difference, but is still often fixed somewhat arbitrarily. For competing risk data, one faces further challenges with univariate parameters, as the absolute risk of a single cause of interest inherently represents a mixture of effects on cause-specific hazards of all competing events. This means that providing inference solely for a single absolute risk probability may lead to obscure conclusions like a medical treatment in fact killing patients displaying a protective effect for the occurrence of events of the cause of interest.

In this work we present methodology for asymptotically linear and efficient substitution estimation of multi-dimensional parameters for general survival and competing risk settings. The observed data are given as $n \in \mathbb{N}$ independent and identically distributed observations of right-censored event times and event indicators taking values in $\mathbb{R}_+ \times \{1, \ldots, J\}$, with each observation drawn from a distribution P_0 assumed to belong to a nonparametric statistical model $\mathcal{M}$. We focus on parameters $\Psi : \mathcal{M} \to \mathscr{H}$ with values in a Hilbert space $\mathscr{H}$ being either a subspace of $\mathbb{R}^d$ for $d \in \mathbb{N}$ or the space of real-valued functions on $\mathbb{R}_+$. Our interest is in the effects of treatment decisions, represented by the special cases of $\Psi : \mathcal{M} \to \mathscr{H}$ corresponding to multivariate effects on state occupation probabilities or effects on an absolute risk or the survival curve across time. Under causal assumptions of consistency, no unmeasured confounding and positivity, our parameters can be interpreted causally as the effects we would see, had we in fact intervened and changed treatment for all subjects in our population. In competing risk settings our methods can be used to make nonparametric inference for the effects on multiple absolute risks across multiple time-points simultaneously, allowing for complete analysis of the effects of a given treatment. More generally, our methods provide a covariate-adjusted alternative to Kaplan–Meier (Kaplan & Meier, 1958) or Aalen-Johansen (Aalen & Johansen, 1978) estimation of survival or absolute risk curves, applicable also in settings where treatment is not randomized and censoring is not independent.

Asymptotically linear and efficient estimation relies on nuisance parameter estimation such as to solve the efficient influence curve equation. In corresponding univariate settings, where parameters take values in $\mathscr{H} \subseteq \mathbb{R}$, several different methods have been proposed to accomplish this (e.g., Hubbard et al., 2000; Moore & van der Laan, 2009a; Stitelman et al., 2011; Benkeser et al., 2018; Ozenne et al., 2020; Rytgaard et al., 2023). None of these methods yields an actual substitution estimator if applied separately for all univariate components of the general multi-dimensional $\Psi : \mathcal{M} \to \mathscr{H}$, and are thus, for example, not guaranteed to respect the parameter space constraints. Furthermore, the majority of these methods rely on a discrete scale for event times and require artificial coarsening to be applied for studies with long follow-up and daily measurements, affecting the very definition of the target parameter and likely causing bias (Sofrygin et al., 2019; Ferreira Guerra et al., 2020).

In this work, we proceed on the basis of one-step targeted maximum likelihood estimation and the concept of universal least favourable submodels (van der Laan & Gruber, 2016; van der Laan & Rose, 2018). Targeted maximum likelihood estimation (van der Laan & Rubin, 2006; van der Laan & Rose, 2011) is a general methodology for semiparametric efficient substitution estimation of causal parameters; the general procedure consists of two steps, where first nuisance parameter estimators are constructed and next updated in a targeted way to solve the efficient influence curve equation. Previous work on targeted maximum likelihood estimation for the considered survival and competing risk settings (Rytgaard et al., 2023) focuses on univariate causal parameters, proposing an iterative targeting procedure that updates cause-specific hazards recursively such as to solve the efficient influence curve equation for that particular univariate parameter. In the present work, we define a universal least favourable submodel for the cause-specific hazards and a corresponding one-step targeting procedure that allows us to solve multiple score equations simultaneously.

We implement our algorithm for one-step targeted maximum likelihood estimation by updating along a corresponding nonuniversal multivariate least favourable submodel. We show that this algorithm can be used to get inference both for multivariate target parameters and target parameters that are real-valued functions on $\mathbb{R}_+$. In the latter case our resulting estimators achieve uniform efficiency and converge weakly to the true absolute risk or survival curve across time. Furthermore, our estimators are true substitution estimators and are thus completely respecting the parameter space constraints of the problem, guaranteed to yield monotone survival curves and state occupation probabilities that sum to one. The simultaneous inference we provide yields a direct methodology for multiple testing correction and also allows us to test for overall effects of treatment. In particular, we can infer on any smooth contrast of the survival curves, such as the restricted mean survival time or the log-rank test statistic (Moore & van der Laan, 2009b). In randomized trial settings where Kaplan–Meier curves stratified by the treatment arm are often presented together with multivariate Cox regression models, our methods provide a unified alternative to take into account covariate information without relying on strong modelling assumptions.

Additionally to solving the efficient influence curve equation, asymptotically linear and efficient estimation relies on nuisance parameter estimation so as to further achieve asymptotic negligibility of an empirical process term and of a second-order remainder (van der Laan & Rubin, 2006; van der Laan & Rose, 2011; van der Laan, 2017). This requires fast enough convergence rates for the estimators as well as small enough function classes assumed for the nuisance parameters contained in model $\mathcal{M}$. Applying highly adaptive lasso estimation (Benkeser & van der Laan, 2016; van der Laan, 2017; Bibaut & van der Laan, 2019), as extended to the present setting (Rytgaard et al., 2022, 2023), for nuisance parameter estimation yields the desired results, relying only on the restriction on $\mathcal{M}$ that nuisance parameters are càdlàg and have finite sectional variation norm (Gill et al., 1995; van der Laan, 2017). Coupling our proposed targeting methodology, which ensures that we solve all relevant efficient influence curve equations, with highly adaptive lasso estimation then produces estimators that achieve asymptotic linearity and efficiency.

2. Setting and notation

We consider a competing risk setting with $J \geqslant 1$ causes corresponding to observed data of the form

$$O = (L, A, \tilde{T}, \tilde{\Delta}) \in \mathbb{R}^{d'} \times \{0, 1\} \times \mathbb{R}_+ \times \{0, 1, \ldots, J\},$$

where $A \in \{0, 1\}$ is a baseline treatment variable, $L \in \mathbb{R}^{d'}$ are pretreatment covariates, $\tilde{T} \in \mathbb{R}_+$ is the observed time under observation and $\tilde{\Delta} \in \{0, 1, \ldots, J\}$ is an indicator of right censoring, $\tilde{\Delta} = 0$, or type of event, $\tilde{\Delta} \geqslant 1$, observed. Variables $T \in \mathbb{R}_+$ and $C \in \mathbb{R}_+$ represent the times to event, one of $J \geqslant 1$ types, and censoring, respectively, such that the observed time is $\tilde{T} = \min(T, C)$ and $\tilde{\Delta} = \mathbb{1}\{T \leqslant C\}\Delta$, where $\Delta \in \{1, \ldots, J\}$ is the event indicator. The special case with $J = 1$ corresponds to a classical survival analysis setting. We let P_0 denote the distribution of O and assume that P_0 belongs to a nonparametric statistical model $\mathcal{M}$. Our data consist of $n \in \mathbb{N}$ observations $O_1, \ldots, O_n$ with each O_i drawn from P_0. Throughout, we use $\mathbb{P}_n$ to denote the empirical distribution of $\{O_i\}_{i=1}^n$.

For $j = 1, \ldots, J$, $\lambda_{0,j}$ denotes the cause j specific hazard, defined as

$$\lambda_{0,j}(t \mid a, \ell) = \lim_{h \to 0} h^{-1} P(T \leqslant t + h,\ \Delta = j \mid T \geqslant t,\ A = a, L = \ell)$$

with $\Lambda_{0,j}(t \mid a, \ell)$ the corresponding cumulative hazard. Likewise, $\lambda_0^c(t \mid a, \ell)$ denotes the conditional hazard for censoring and $\Lambda_0^c(t \mid a, \ell)$ the corresponding cumulative hazard. We further let μ_0 be the density of L with respect to an appropriate dominating measure ν and $\pi_0(\cdot \mid L)$ the conditional distribution of A given L. The distribution for the observed data can now be represented as $dP_0(o) = p_0(o)\, d\nu(\ell)\, dt$, where $o = (\ell, a, t, \delta)$ and the density p_0 under coarsening at random (van der Laan & Robins, 2003) factorizes as

$$p_0(o) = \mu_0(\ell)\pi_0(a \mid \ell)\{\lambda_0^c(t \mid a, \ell)\}^{\mathbb{1}\{\delta=0\}} S_0^c(t- \mid a, \ell) \prod_{j=1}^{J} \{\lambda_{0,j}(t \mid a, \ell)\}^{\mathbb{1}\{\delta=j\}} S_0(t- \mid a, \ell);$$

here, still under coarsening at random, $S_0(t \mid a, \ell) = \exp\{-\int_0^t \sum_{j=1}^J \lambda_{0,j}(s \mid a, \ell)\, \mathrm{d}s\}$ denotes the survival function and $S_0^c(t \mid a, \ell) = \exp\{-\int_0^t \lambda_0^c(s \mid a, \ell)\, \mathrm{d}s\}$ the censoring survival function. For $j = 1, \ldots, J$, we further denote by

$$F_{0,j}(t \mid a, \ell) = \int_0^t S_0(s- \mid a, \ell)\lambda_{0,j}(s \mid a, \ell)\, \mathrm{d}s$$

the absolute risk function, or the subdistribution, for events of type j (Gray, 1988).

3. Target of estimation

3.1. *Univariate statistical parameter and its causal interpretation*

In this paper we define target parameters that are multi-dimensional along two dimensions: across types of events and across time. This will allow us to evaluate treatment effects on the risk of occurrences of multiple competing events simultaneously as well as to evaluate treatment effects as they vary across time. Here we first define the univariate components of our multi-dimensional parameters of interest. Let $\Psi_{j,t} \colon \mathcal{M} \to \mathbb{R}$ be the univariate parameter given by

$$\Psi_{j,t}(P) = \mathbb{E}\{F_j(t \mid A = a^*, L)\} = \mathbb{E}\{P(T \leqslant t,\ \Delta = j \mid A = a^*, L)\} \tag{1}$$

for fixed cause j and time-point $t \in (0, \tau]$. Here a^* could be either 1 or 0, representing the treatment- or control-specific probabilities. Taking the difference between the two corresponds to the average treatment effect. Furthermore, when there are no competing risks, (1) is simply one minus the treatment-specific survival curve.

Causal assumptions of consistency, no unmeasured confounding and positivity yield a causal interpretation of (1). These assumptions involve the counterfactual event time T^{a^*} and the counterfactual event indicator Δ^{a^*} defined as the event time and event indicator we would have observed for a subject had they been treated according to the treatment level

$A = a^*$, irrespective of what treatment they in fact received (Hernan & Robins, 2020). Consistency is the assumption that $T = T^{a^*}$ and $\Delta = \Delta^{a^*}$ on the event that $A = a^*$ for $a^* = 0, 1$, i.e., that the observed event time and indicator are equal to the counterfactual ones under the treatment level observed. No unmeasured confounding is needed in two regards. First, conditionally independent censoring, i.e., the assumption that $(T, \Delta) \perp\!\!\!\perp C \mid A, L$, identifies (1) as the absolute risk probability standardized with respect to the distribution of observed covariates. Second, no unmeasured confounding on the treatment decisions, i.e., the assumption that $(T^{a^*}, \Delta^{a^*}) \perp\!\!\!\perp A \mid L$, yields the interpretation of (1) as the absolute risk of events of type j before time t we would observe had in fact all subjects of the population been assigned treatment level $A = a^*$. The assumptions together thus imply that

$$\Psi_{j,t}(P) = P(T^{a^*} \leqslant t,\ \Delta^{a^*} = j). \tag{2}$$

Finally, throughout, we make the positivity assumption that $S^c(t- \mid a^*, L)\pi(a^* \mid L) > \eta > 0$, μ-almost everywhere, which is necessary for the parameter defined by (1) to be well defined as a statistical parameter. The formal proof of (2) can be found in the Supplementary Material.

3.2. *Multi-dimensional parameters of interest*

In this paper we are concerned mainly with two different multi-dimensional parameters with components given by (1) across causes and time.

In competing risk settings, it is well known that a treatment effect on the absolute risk of a single event type of interest may be explained partially by its effect on competing risk events. The first parameter will allow us to assess treatment effects on multiple absolute risk simultaneously, possibly also over a range of time-points. We define the parameter as the multivariate mapping $\Psi^{\mathrm{mv}}\colon \mathcal{M} \to \mathscr{H}_{\Sigma_d} \subset \mathbb{R}^d$ given by

$$\Psi^{\mathrm{mv}}(P) = \{\Psi_{j,t_k}(P)\colon j = 1, \ldots, J,\ k = 1, \ldots, K\},$$

where $d = JK$ and $\mathscr{H}_{\Sigma_d}$ is the Hilbert space of elements $x \in \mathbb{R}^d$ endowed with inner product and corresponding norm

$$\langle x, y\rangle_{\Sigma_d} = x^{\mathrm{T}}\Sigma_d^{-1}y, \qquad \|x\|_{\Sigma_d} = (x^{\mathrm{T}}\Sigma_d^{-1}x)^{1/2}, \tag{3}$$

for a user-supplied positive definite matrix $\Sigma_d \in \mathbb{R}^d \times \mathbb{R}^d$. Matrix Σ_d could simply be the identity matrix, so that the norm in (3) is the standard Euclidean norm, but later we consider and discuss other options as well.

The other parameter we are interested in is the treatment-specific average cause-j-specific absolute risk, or survival, curve across all time-points in the continuum of the follow-up period. We define this as the infinite-dimensional parameter $\Psi_j^{\mathrm{inf}}\colon \mathcal{M} \to \mathscr{H}_{[0,\tau]}$ given by

$$\Psi^{\mathrm{inf}}(P) = \{\Psi_{j,t}(P)\colon t \in [0, \tau]\}$$

with values in the Hilbert space $\mathscr{H}_{[0,\tau]}$ of real-valued functions on $\mathbb{R}_+$ endowed with the inner product

$$\langle f_1, f_2\rangle_{\mathscr{S}} = \int_0^\tau f_1(t)\mathscr{S}f_2(t)\,\mathrm{d}t = \int_0^\tau f_1(t)\int_0^\tau \mathcal{S}(s,t)f_2(s)\,\mathrm{d}s\,\mathrm{d}t$$

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for a user-supplied positive definite operator $\mathscr{S}$ with kernel function $\mathcal{S}(s,t)$. The operator can be chosen such that the finite-dimensional evaluation over time-points $t_1,\dots,t_K$ becomes the matrix inner product corresponding to (3). The simple kernel function $\mathcal{S}(s,t)=1$ corresponds to Σ_d chosen as the identity matrix.

4. Overview of estimation

4.1. *Efficient influence function*

In § 5 we present our estimation procedure for asymptotically linear and efficient substitution estimation of the multi-dimensional parameters from § 3.2. Here we present the efficient influence function that enters directly in this estimation procedure and completely characterizes the asymptotic distributions of the resulting estimators. We summarize in § 4.2 the procedure of targeted estimation and in § 4.3 we present the asymptotic distribution of the estimators that we achieve with our estimation procedure.

The efficient influence function of the (j,t)-specific univariate parameter $\Psi_{j,t}$ at P relative to $\mathcal{M}$ is given by

$$D^*_{j,t}(P)(O)=\sum_{l=1}^{J}\int h_{j,l,t,s}(P)(O)[N_l(ds)-\mathbb{1}\{\tilde{T}\geqslant s\}\lambda_l(s\mid A,L)\,\mathrm{d}s] + F_j(t\mid a,L)-\Psi_{j,t}(P)$$

with $N_l(t)=\mathbb{1}\{\tilde{T}\leqslant t,\ \Delta=l\}$ for $l=1,\dots,J$ denoting the observed counting processes and the functions $h_{j,l,t,s}$ defined by

$$h_{j,l,t,s}(P)(O)=\frac{\mathbb{1}\{A=a\}}{\pi(A\mid L)}\frac{\mathbb{1}\{s\leqslant t\}}{S^c(s-\mid A,L)} \times\begin{cases}1-\dfrac{F_l(t\mid A,L)-F_l(s\mid A,L)}{S(s\mid A,L)} & \text{when } l=j,\\ -\dfrac{F_j(t\mid A,L)-F_j(s\mid A,L)}{S(s\mid A,L)} & \text{when } l\neq j.\end{cases} \tag{4}$$

The derivation can be found in the Supplementary Material and also, for $J=2$, in the supplementary material of Rytgaard et al. (2023). As we shall see in § 5, the functions defined by (4) characterize what we refer to as the least favourable path for constructing our estimators.

Whereas the parameter defined by (1), which can also be written as

$$\Psi_{j,t}(P)=\int_{\mathcal{L}}F_j(t\mid a^*,\ell)\mu(\ell)\,\mathrm{d}\nu(\ell)=\int_{\mathcal{L}}\int_0^t S(s-\mid a^*,\ell)\Lambda_j(\mathrm{d}s\mid a^*,\ell)\mu(\ell)\,\mathrm{d}\nu(\ell),$$

depends on P through the covariate density μ and all cause-specific hazards $\lambda=(\lambda_l\colon l=1,\dots,J)$ via $S(t\mid a^*,\ell)=\exp\{-\int_0^t\sum_{l=1}^J\lambda_l(s\mid a^*,\ell)\,\mathrm{d}s\}$, the efficient influence function also depends on the treatment propensity π and the censoring survival function S^c. To reflect these dependencies on different parts of the data-generating distribution in our notation, we use the alternative notation

$$\tilde{\Psi}_{j,t}(\lambda):=\Psi_{j,t}(P),\qquad \tilde{D}^*_{j,t}(\lambda,\pi,S^c):=D^*_{j,t}(P)\quad\text{and}\quad \tilde{h}_{j,l,t,s}(\lambda,\pi,S^c):=h_{j,l,t,s}(P),$$

when appropriate. We have suppressed the dependence on μ in this notation, as we simply estimate the average over the covariate distribution by the empirical average.

We further introduce the vectorized notation

$$\tilde{D}^* = (\tilde{D}^*_{j,t_k} : j = 1, \ldots, J,\ k = 1, \ldots, K), \tag{5}$$

$$\tilde{h}_{l,s} = (\tilde{h}_{j,l,t_k,s} : j = 1, \ldots, J,\ k = 1, \ldots, K), \tag{6}$$

to refer to the d-dimensional vector of stacked efficient influence functions and functions indexing the least favourable paths, respectively. With this notation, $\tilde{D}^*$ is the efficient influence function for the multivariate parameter $\Psi^{\mathrm{mv}} \colon \mathcal{M} \to \mathscr{H}_{\Sigma_d}$.

4.2. *Targeted estimation*

Asymptotically linear and efficient estimation of the multi-dimensional parameters from § 3.2 relies on nuisance parameter estimators solving the efficient influence curve equation and also converging to their true counterparts at a fast enough rate. Targeted estimation generally proceeds in two steps to achieve this, with an initial estimation step and a targeting step. In the first step estimators $\hat{\lambda}_n = (\hat{\lambda}_{l,n} \colon l = 1, \ldots, J)$, $\hat{\pi}_n$ and $\hat{S}^c_n$ are constructed, being initial in the sense that they are not yet required to solve any efficient influence curve equation. As their construction can rely entirely on existing work, here we simply assume that we have at hand these initial estimators so that we can focus fully on the targeting step.

In the targeting step, an updated version $\hat{\lambda}^*_n = (\hat{\lambda}^*_{l,n} \colon l = 1, \ldots, J)$ is constructed such that $\hat{\lambda}^*_n$, $\hat{\pi}_n$ and $\hat{S}^c_n$ together solve the efficient influence curve equation. For the (j, t)-specific parameter, this means that

$$\mathbb{P}_n \tilde{D}^*_{j,t}(\hat{\lambda}^*_n, \hat{\pi}_n, \hat{S}^c_n) = o_P(n^{-1/2}), \tag{7}$$

whereas for the multivariate and the infinite-dimensional parameters, it amounts to solving multiple equations across j and t simultaneously. The standard procedure for the targeting step (van der Laan & Rubin, 2006; van der Laan & Rose, 2011) uses the ‘least favourable’, or ‘hardest’, submodel (van der Vaart, 2000) as a fluctuation model to iteratively update a current estimator. Per definition, the least favourable submodel is the parametric model $(\lambda_\varepsilon \colon \varepsilon \in \mathbb{R})$ that at $\varepsilon = 0$ optimizes the Cramér–Rao lower bound for estimating $\Psi(P_0)$ given model $\mathcal{M}$. Updating along this submodel proceeds with maximum likelihood estimation steps and usually requires iteration to reach the point at which the efficient influence curve equation is solved. If we only cared about univariate inference, we could apply the iterative targeting procedure proposed by Rytgaard et al. (2023). However, using this for the multi-dimensional parameters would result, for each (j, t)-specific equation needed to be solved, in a new estimator for λ used to estimate the target parameter for that j and t. In § 5 we proceed instead with a one-step targeting procedure, updating along a universal least favourable submodel (van der Laan & Gruber, 2016; van der Laan & Rose, 2018, Ch. 5) $(\lambda_\varepsilon \colon \varepsilon \in \mathbb{R})$ defined so that it reaches the Cramér–Rao bound for estimating $\Psi(P_0)$ given model $\mathcal{M}$ at any amount of fluctuation ε. With this procedure, we can update estimators in a single step and solve the efficient influence curve equations across all j and t simultaneously.

4.3. *Asymptotic distribution of targeted estimators*

After applying the targeting procedure that we describe in § 5, each univariate target parameter is estimated by $\hat{\psi}^*_{n,j,t} = \tilde{\Psi}_{j,t}(\hat{\lambda}^*_n)$. By virtue of solving the efficient influence curve equation, and with the initial estimators meeting the required conditions reviewed in § 1, this estimator achieves asymptotic linearity and efficiency. In particular, when estimators $\hat{\lambda}_n^*$, $\hat{\pi}_n$ and $\hat{S}_n^c$ solve the efficient influence curve equation (7) across all $j = 1, \ldots, J$ and $k = 1, \ldots, K$, estimator $\hat{\psi}_n^{\mathrm{mv},*} = (\hat{\psi}^*_{n,j,t_k} : j = 1, \ldots, J,\ k = 1, \ldots, K)$ for $\psi_0^{\mathrm{mv}} = (\psi_{0,j,t_k} : j = 1, \ldots, J,\ k = 1, \ldots, K)$ follows the asymptotic distribution

$$\surd n(\hat{\psi}_n^{\mathrm{mv},*} - \psi_0^{\mathrm{mv}}) \xrightarrow{\mathrm{D}} N(0, \Sigma_0) \tag{8}$$

with '$\xrightarrow{\mathrm{D}}$' denoting convergence in distribution and $\Sigma_0 = D^*(P_0)D^*(P_0)^{\mathrm{T}} \in \mathbb{R}^d \times \mathbb{R}^d$ being the covariance matrix of the stacked efficient influence function $\tilde{D}^* = (\tilde{D}^*_{j,t_k} : j = 1, \ldots, J,\ k = 1, \ldots, K)$. Similarly, when estimators $\hat{\lambda}_n^*$, $\hat{\pi}_n$ and $\hat{S}_n^c$ solve the efficient influence curve equation (7) uniformly across all $t \in [0, \tau]$, we obtain the asymptotic distribution of estimator $\hat{\psi}_{n,j}^{\mathrm{inf},*} = (\hat{\psi}^*_{n,j,t} : t \in [0, \tau])$ for $\psi_{0,j}^{\mathrm{inf}} = (\psi_{0,j,t} : t \in [0, \tau])$,

$$\surd n(\hat{\psi}_{n,j}^{\mathrm{inf},*} - \psi_{0,j}^{\mathrm{inf}}) = \surd n(\mathbb{P}_n - P_0)D^*(P_0) + o_P(1) \xrightarrow{\mathrm{D}} \mathbb{G}_0,$$

where $\mathbb{G}_0$ is a Gaussian process with covariance structure given by the covariance operator $\mathcal{S}(t_1, t_2) = P_0 D^*_{j,t_1}(P_0) D^*_{j,t_2}(P_0)$. Specifically, estimator $\hat{\psi}_{n,j}^{\mathrm{inf},*}$ achieves uniform asymptotic efficiency and converges weakly to the true function.

5. One-step targeted maximum likelihood estimation

5.1. *Overview*

We now present our one-step targeted maximum likelihood estimation procedure applied to the initial estimators $\hat{\lambda}_n = (\hat{\lambda}_{l,n} : l = 1, \ldots, J)$, $\hat{\pi}_n$ and $\hat{S}_n^c$ to update $\hat{\lambda}_n$ into $\hat{\lambda}_n^*$, such that $\hat{\lambda}_n^*$, $\hat{\pi}_n$ and $\hat{S}_n^c$ together solve the efficient influence curve equation (7) across causes $j = 1, \ldots, J$ and a range of time-points $t_1, \ldots, t_K \in [0, \tau]$. The resulting substitution estimator $\hat{\psi}_n^{\mathrm{mv},*} = (\Psi^{\mathrm{mv}}(\hat{\lambda}_n^*) : j = 1, \ldots, J,\ k = 1, \ldots, K)$ for the multivariate parameter $\Psi^{\mathrm{mv}} : \mathcal{M} \to \mathscr{H}_{\Sigma_d}$ is then asymptotically linear and efficient and follows the asymptotic distribution (8). The one-step targeted maximum likelihood estimation procedure we propose revolves around a universal least favourable submodel for the cause-specific hazards that we define in § 5.2. In § 5.3 we describe our algorithm to find the maximum likelihood estimator along this universal least favourable submodel. In § 5.4 we discuss variations over the Hilbert space norms where our parameters and estimators take value. In § 5.5 we show that our algorithm not only solves the efficient influence curve equation over a finite range of time-points, but also across all time-points in the continuum $[0, \tau]$. This will allow us to construct asymptotically linear and efficient substitution estimators for the infinite-dimensional parameter $\Psi_j^{\mathrm{inf}} : \mathcal{M} \to \mathscr{H}_{[0,\tau]}$ that converges weakly to the true treatment-specific absolute risk curve.

5.2. *Universal least favourable submodel*

Below we define a universal least favourable submodel $\lambda_\varepsilon = (\lambda_{l,\varepsilon} : l = 1, \ldots, J)$ for $\lambda = (\lambda_l : l = 1, \ldots, J)$ such that, for any $\varepsilon \in \mathbb{R}$, the loglikelihood loss function $(O, \lambda) \mapsto \mathscr{L}(\lambda)(O)$ and, for fixed π and S^c, we have

$$\frac{d}{d\varepsilon} \mathbb{P}_n \mathscr{L}(\lambda_\varepsilon) = \| \mathbb{P}_n \tilde{D}^*(\lambda_\varepsilon, \pi, S^c) \|_{\Sigma_d}. \tag{9}$$

Here and in the following we use the vectorized notation from (5)–(6). Denoting by $\hat{\lambda}_{n,\varepsilon} = (\hat{\lambda}_{l,n,\varepsilon} : l = 1, \ldots, J)$ the universal least favourable submodel applied to our initial estimator, the defining property (9) implies that the maximum likelihood estimator along the path defined by the universal least favourable submodel,

$$\hat{\varepsilon}_n = \arg\min_{\varepsilon \in \mathbb{R}} \mathbb{P}_n \mathscr{L}(\hat{\lambda}_{n,\varepsilon}), \tag{10}$$

is a local maximum. Thus, estimator $\hat{\lambda}_{n,\hat{\varepsilon}_n}$ solves the equation

$$\|\mathbb{P}_n \tilde{D}^*(\hat{\lambda}_{n,\hat{\varepsilon}_n}, \hat{\pi}_n, \hat{S}^c_n)\|_{\Sigma_d} = 0,$$

and it is then further a straightforward consequence that $\mathbb{P}_n \tilde{D}^*_{j,t_k}(\hat{\lambda}_{n,\hat{\varepsilon}_n}, \hat{\pi}_n, \hat{S}^c_n) = 0$ for every $j = 1, \ldots, J$ and $k = 1, \ldots, K$, i.e., all desired efficient influence curve equations are solved simultaneously.

Definition 1 (Universal least favourable submodel). *For each cause-specific hazard, we define*

$$\lambda_{l,\varepsilon}(s) = \lambda_l(s) \exp\left\{ \int_0^{\varepsilon} \frac{\langle \mathbb{P}_n \tilde{D}^*(\lambda_x, \pi, S^c), \tilde{h}_{l,s}(\lambda_x, \pi, S^c)\rangle_{\Sigma_d}}{\|\mathbb{P}_n \tilde{D}^*(\lambda_x, \pi, S^c)\|_{\Sigma_d}}\, dx \right\} \qquad (l = 1, \ldots, J),$$

and refer to the collection $\lambda_\varepsilon = (\lambda_{l,\varepsilon} : l = 1, \ldots, J)$ as the universal least favourable submodel.

Let $(O, \lambda) \mapsto \mathscr{L}(\lambda)(O)$ denote the sum loglikelihood loss function given as

$$\mathscr{L}(\lambda)(O) = -\sum_{l=1}^{J} \left\{ \int_0^{\tau} \log \lambda_l(s \mid A, L) N(\mathrm{d}s) - \int_0^{\tau} \mathbb{1}\{\tilde{T} \geqslant s\} \lambda_l(s \mid A, L)\, \mathrm{d}s \right\}. \tag{11}$$

For this loss function and the universal least favourable submodel from Definition 1, we indeed have

$$\begin{aligned}
\frac{d}{d\varepsilon} \mathbb{P}_n \mathscr{L}(\lambda_\varepsilon) &= \frac{\{\mathbb{P}_n \tilde{D}^*(\lambda_\varepsilon, \pi, S^c)\}^{\mathrm{T}} \Sigma_d^{-1}}{\|\mathbb{P}_n \tilde{D}^*(\lambda_\varepsilon, \pi, S^c)\|_{\Sigma_d}} \mathbb{P}_n \left[\sum_{l=1}^{J} \int_0^{\tau} \tilde{h}_{l,s}(\lambda_\varepsilon, \pi, S^c) \{N_l(ds) - \lambda_{l,\varepsilon}(s)\, \mathrm{d}s\} \right] \\
&= \frac{\{\mathbb{P}_n \tilde{D}^*(\lambda_\varepsilon, \pi, S^c)\}^{\mathrm{T}} \Sigma_d^{-1}}{\|\mathbb{P}_n \tilde{D}^*(\lambda_\varepsilon, \pi, S^c)\|_{\Sigma_d}} \mathbb{P}_n \tilde{D}^*(\lambda_\varepsilon, \pi, S^c) \\
&= \|\mathbb{P}_n \tilde{D}^*(\lambda_\varepsilon, \pi, S^c)\|_{\Sigma_d},
\end{aligned}$$

which verifies the desired property (9).

5.3. *Recursive implementation of the one-step targeting procedure*

We implement our one-step targeting algorithm for finding the maximum likelihood estimator (10) by a recursive implementation updating any current estimator locally along a nonuniversal, or local, least favourable submodel with a multivariate parameter. This follows the practical construction suggested by van der Laan & Rose (2018, § 5.5.1), adapted to our setting.

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For our implementation, we define a multivariate least favourable submodel with parameter $\delta \in \mathbb{R}^d$ as follows:

$$\lambda_\delta^{\mathrm{lfm}}(s) = \Big[\lambda_l(s) \exp\{\langle \tilde{h}_{l,s}(\lambda, \pi, S^c), \delta\rangle_{\Sigma_d}\} \colon l = 1, \ldots, J\Big]. \tag{12}$$

With the loss function defined by (11), this submodel fulfills

$$\left\{ \frac{d}{d\delta}\bigg|_{\delta=0} \mathscr{L}(\lambda_\delta^{\mathrm{lfm}}) \right\}^{\mathrm{T}} \delta = \langle \tilde{D}^*(\lambda, \pi, S^c), \delta\rangle_{\Sigma_d},$$

i.e., the linear span of its score at $\delta = 0$ includes the efficient influence function. This is exactly the defining property of a local least favourable submodel. Maximizing the loglikelihood $\mathbb{P}_n\mathscr{L}(\lambda_\delta^{\mathrm{lfm}})$ locally corresponds to maximizing the gradient approximation $\langle \mathbb{P}_n\tilde{D}^*(\lambda, \pi, S^c), \delta\rangle_{\Sigma_d}$. This can bounded by the Cauchy-Schwarz inequality

$$\langle \mathbb{P}_n\tilde{D}^*(\lambda, \pi, S^c), \delta\rangle_{\Sigma_d} \leqslant \|\mathbb{P}_n\tilde{D}^*(\lambda, \pi, S^c)\|_{\Sigma_d} \|\delta\|_{\Sigma_d},$$

and thus maximized over δ locally with $\|\delta\|_{\Sigma_d} \leqslant dx$ for a small number $dx \geqslant 0$ by

$$\delta_n^* = \frac{\mathbb{P}_n\tilde{D}^*(\lambda, \pi, S^c)}{\|\mathbb{P}_n\tilde{D}^*(\lambda, \pi, S^c)\|_{\Sigma_d}}\, dx.$$

Evaluating the multivariate least favourable submodel (12) in the locally approximate optimal parameter δ_n^* now gives

$$\lambda_{\delta_n^*}(s) = \left[\lambda_l(s) \exp\left\{ \frac{\langle \mathbb{P}_n\tilde{D}^*(\lambda, \pi, S^c), \tilde{h}_{l,s}(\lambda, \pi, S^c)\rangle_{\Sigma_d}}{\|\mathbb{P}_n\tilde{D}^*(\lambda, \pi, S^c)\|_{\Sigma_d}}\, dx \right\} \colon l = 1, \ldots, J\right]. \tag{13}$$

A recursive updating scheme where a current λ_x is updated in the locally least favourable direction as defined by (13) with λ_x substituted for λ yields the universal least favourable submodel. Thus, specifically, with $dx \geqslant 0$ denoting a small step size, we define, for each $l = 1, \ldots, J$,

$$\hat{\lambda}_{l,n}^{(1)}(s) = \hat{\lambda}_{l,n}(s) \exp\left\{ \frac{\langle \mathbb{P}_n\tilde{D}^*(\hat{\lambda}_n, \hat{\pi}_n, \hat{S}_n^c), \tilde{h}_{l,s}(\hat{\lambda}_n, \hat{\pi}_n, \hat{S}_n^c)\rangle_{\Sigma_d}}{\|\mathbb{P}_n\tilde{D}^*(\hat{\lambda}_n, \hat{\pi}_n, \hat{S}_n^c)\|_{\Sigma_d}}\, dx \right\}$$

and, recursively for $m \geqslant 1$,

$$\hat{\lambda}_{l,n}^{(m+1)}(s) = \hat{\lambda}_{l,n}^{(m)}(s) \exp\left\{ \frac{\langle \mathbb{P}_n\tilde{D}^*(\hat{\lambda}_n^{(m)}, \hat{\pi}_n, \hat{S}_n^c), \tilde{h}_{l,s}(\hat{\lambda}_n^{(m)}, \hat{\pi}_n, \hat{S}_n^c)\rangle_{\Sigma_d}}{\|\mathbb{P}_n\tilde{D}^*(\hat{\lambda}_n^{(m)}, \hat{\pi}_n, \hat{S}_n^c)\|_{\Sigma_d}}\, dx \right\}.$$

Our algorithm is summarized as follows.

Algorithm 1. One-step targeting procedure for $\Psi\colon \mathcal{M} \to \mathscr{H}_{\Sigma_d}$.

Initialize estimators $\hat\lambda_n^{(0)} \leftarrow (\hat\lambda_{l,n}\colon l = 1,\ldots,J)$, $\hat S_n$ and $\hat\pi_n$.
for $m \leftarrow 0, 1, \ldots$ do
  evaluate $\hat\lambda_{l,n}^{(m+1)}(s) \leftarrow \lambda_{l,n}^{(m)}(s)\exp\left(\frac{\langle \mathbb{P}_n \tilde D^*(\hat\lambda_n^{(m)},\hat\pi_n,\hat S_n^c), \tilde h_{l,s}(\hat\lambda_n^{(m)},\hat\pi_n,\hat S_n^c)\rangle_{\Sigma_d}}{\|\mathbb{P}_n\tilde D^*(\hat\lambda_n^{(m)},\hat\pi_n,\hat S_n^c)\|_{\Sigma_d}}\cdot dx\right)$;
  if $\|\mathbb{P}_n\tilde D^*(\hat\lambda_n^{(m)},\hat\pi_n,\hat S_n^c)\|_{\Sigma_d} \leqslant \|\mathbb{P}_n\tilde D^*(\hat\lambda_n^{(m+1)},\hat\pi_n,\hat S_n^c)\|_{\Sigma_d}$ then
    $dx \leftarrow 0.5\cdot dx$;
    $m \leftarrow m-1$;
    break
  end
  evaluate $s_n^{(m)} \leftarrow \max_{j,k} \frac{|\mathbb{P}_n\tilde D^*_{j,t_k}(\hat\lambda_n^{(m+1)},\hat\pi_n,\hat S_n^c)|}{\surd\mathbb{P}_n\tilde D^*_{j,t_k}(\hat\lambda_n^{(m+1)},\hat\pi_n,\hat S_n^c)^2}$;
  if $s_n^{(m)} \leqslant 1/(\surd n\log n)$ then
    break
  end
end
evaluate $\hat\lambda_n^* \leftarrow \hat\lambda_n^{(m+1)}$;
evaluate $\hat\psi_n^* \leftarrow \tilde\Psi(\hat\lambda_n^*)$.

Recall again that $\hat\lambda_{n,\varepsilon} = (\hat\lambda_{l,n,\varepsilon}\colon l = 1,\ldots,J)$ is the universal least favourable submodel applied to our initial estimator. Estimator $\hat\lambda_n^*$ achieved with Algorithm 1 corresponds to $\hat\lambda_{n,\hat\varepsilon_n}$, where $\hat\varepsilon_n$ is the maximum likelihood estimator (10) along the path defined by the universal least favourable submodel. Now, Algorithm 1 requires that we can control the maximum norm, i.e., that there is in fact an $\hat\varepsilon_n$ such that

$$\max_{j,k} \frac{|\mathbb{P}_n\tilde D^*_{j,t_k}(\hat\lambda_{n,\hat\varepsilon_n},\hat\pi_n,\hat S_n^c)|}{\surd\mathbb{P}_n\tilde D^*_{j,t_k}(\hat\lambda_{n,\hat\varepsilon_n},\hat\pi_n,\hat S_n^c)^2} \leqslant \frac{1}{\surd n\log n};$$

Lemma 1 in the Supplementary Material tells us that we can achieve exactly this.

5.4. *Variations over the Hilbert space norm*

We noted in § 3.2 that matrix $\Sigma_d \in \mathbb{R}^d \times \mathbb{R}^d$ entering the Hilbert space norm (3) could simply be the identity matrix. Here we discuss two other options where the matrix depends on the efficient influence function. With the multi-dimensional parameters we are interested in, we may see both large variations in variance, such as across different subdistributions, or, particularly when there are positivity issues, across different time-points, but also strong correlations, such as across time-points close to each other. With the following variations over Σ_d, we can take such patterns into account, recalling that the covariance of the efficient influence function equals the asymptotic covariance of the estimator for the target parameter.

(i) *Norm weighted by the variance of the efficient influence function.* We can define $\Sigma_d \in \mathbb{R}^d \times \mathbb{R}^d$ as the diagonal matrix with diagonal values given by the variances $\sigma^2_{j,k} = P_0\{\tilde D^*_{j,k}(\lambda,\pi,S^c)\}^2$ of the efficient influence functions.

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(ii) *Norm weighted by the covariance of the efficient influence function.* We can define $\Sigma_d \in \mathbb{R}^d \times \mathbb{R}^d$ as the covariance matrix $\Sigma_0 = P_0 D^*(\lambda, \pi, S^c) D^*(\lambda, \pi, S^c)^{\mathrm{T}}$ of the stacked efficient influence function $\tilde{D}^*(\lambda, \pi, S^c)$.

In the estimation procedure, we replace Σ_d by its empirical counterpart, with the estimated variances $\hat{\sigma}^2_{j,k} = \mathbb{P}_n\{\tilde{D}^*_{j,k}(\hat{\lambda}_n, \hat{\pi}_n, \hat{S}^c_n)\}^2$ or the empirical covariance matrix $\Sigma_n = \mathbb{P}_n D^*(\hat{\lambda}_n, \hat{\pi}_n, \hat{S}^c_n) D^*(\hat{\lambda}_n, \hat{\pi}_n, \hat{S}^c_n)^{\mathrm{T}}$.

The norm defined by (3) corresponds to the Mahalanobis norm, measuring the weighted distance to zero. When weighting by the variance of the efficient influence function, the individual elements of the efficient influence curve will contribute to $\|\mathbb{P}_n \tilde{D}^*(\hat{\lambda}_{n,\hat{\varepsilon}_n}, \hat{\pi}_n, \hat{S}^c_n)\|_{\Sigma_d}$ in inverse proportion to their variance. This means, for example, that elements of the target parameter for which there is less information will be less harmful to the estimation procedure. When weighting by the covariance matrix, the norm $\|\mathbb{P}_n \tilde{D}^*(\hat{\lambda}_{n,\hat{\varepsilon}_n}, \hat{\pi}_n, \hat{S}^c_n)\|_{\Sigma_d}$ will measure the distance to the distribution of the empirical mean of the efficient influence curve. If the different elements of the efficient influence curve are strongly correlated then the distance will be reduced accordingly.

Although variations over the choice of Σ_d may have substantial finite-sample implications, it does not alter the asymptotic properties of the resulting estimator that is still implemented along the lines of Algorithm 1.

5.5. *One-step targeted estimation of the infinite-dimensional parameter*

Here we establish that the practical construction of the one-step targeting procedure for $\Psi^{\inf}: \mathcal{M} \to \mathscr{H}_{[0,\tau]}$ can in fact be carried out along a grid of time-points $0 \leqslant t_1 < t_2 < \cdots < t_{M_n} \leqslant \tau$ as long as the grid is chosen fine enough. Specifically, this means that the implementation may follow exactly along the lines of Algorithm 1; as established by Theorem 1 below, this will be enough to ensure that the efficient influence curve equation is solved across all $t \in [0, \tau]$. In the end this is what we need to achieve weak convergence of the resulting targeted estimator for the treatment-specific absolute risk curve.

Theorem 1 (Targeting over a grid). *Define a grid* $0 \leqslant t_1 < t_2 < \cdots < t_{M_n} \leqslant \tau$ *of time-points in* $[0, \tau]$ *fine enough such that* $\max_m (t_m - t_{m-1}) = O_P(n^{-1/3-\eta})$ *for some* $\eta > 0$. *For each* $t \in [0, \tau]$, *let* $m(t) := \min_m |t - t_m|$. *Assume that*

(i) $S(\tau \mid A, L) > \kappa'$ *for some* $\kappa' > 0$,
and that, for all $t \in [0, \tau]$,

(ii) $P_0\{D^*_{j,t}(\hat{P}^*_n) - D^*_{j,t_{m(t)}}(\hat{P}^*_n)\}^2 \xrightarrow{\mathrm{P}} 0$,

(iii) $\|(h_{1,\ell,t_{m(t)}} - h_{1,\ell,t})(\hat{P}^*_n)\|_{\mu_0 \otimes \pi_0 \otimes \rho} \leqslant K' |t_{m(t)} - t|^{1/2}$ *for a constant* $K' > 0$; *here,* ρ *denotes the Lebesgue measure and* $\| \cdot \|_{\mu_0 \otimes \pi_0 \otimes \rho}$ *the* $L_2(\mu_0 \otimes \pi_0 \otimes \rho)$*-norm, where* $\mu_0 \otimes \pi_0 \otimes \rho$ *is the product measure of* μ_0, π_0 *and* ρ.

Now, if

$$\max_{t \in \{t_1, \ldots, t_{M_n}\}} \mathbb{P}_n D^*_{j,t}(\hat{P}^*_n) = o_P(n^{-1/2})$$

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then we have

$$\sup_{t\in[0,\tau]} \mathbb{P}_n D^*_{j,t}(\hat{P}^*_n) = o_P(n^{-1/2}). \tag{14}$$

The proof of Theorem 1 and a verification that part (iii) holds under a Lipschitz-type condition on the cause j subdistribution can be found in the Supplementary Material. The exponent of $\alpha = 1/2$ for the upper bound $|t_m - t|^\alpha$ of part (iii) is specific for our parameter of interest. For other choices of target parameters, it may be that $\alpha \in [\frac{1}{2}, 1]$ such that a coarser grid is really needed for weak convergence.

If the grid of time-points is not chosen fine enough, so that there are time-points where the efficient influence curve equation is not solved, this introduces an additional bias term at those time-points specifically. In any given concrete application, however, we emphasize that it can always be checked whether (14) is obtained, and that the size of the finite grid of time-points should, as we demonstrate empirically in the Supplementary Material, be increased until it is the case. Asymptotically, and thus for large sample sizes, this should always be doable. To enhance the practical applicability in finite samples, slight adjustment of our algorithm, such as proposed in the Supplementary Material, may be implemented.

6. Empirical study

6.1. *Demonstration*

Our empirical study consists here of a demonstration of our proposed methodology on a publicly available dataset. In the Supplementary Material we present further simulation studies of the properties of different variations of targeting, including variations over the choice of Hilbert space norm as described in § 5.4.

We consider the dataset on follicular cell lymphoma as available in R from the `rfsrc` package (Ishwaran & Kogalur, 2022; R Development Core Team, 2024). In these data, the treatment variable $A \in \{0, 1\}$ is coded such that $A = 1$ represents a combination treatment of radiation and chemotherapy and $A = 0$ represents treatment with radiation alone. The data also include baseline information on age, clinical disease stage and hemoglobin level and the observed time $\tilde{T}$ to disease relapse, $\tilde{\Delta} = 1$, death without relapse, $\tilde{\Delta} = 2$, or right censoring, $\tilde{\Delta} = 0$. The data display substantial time-varying effects; see, e.g., Scheike & Zhang (2011). Here we consider estimation of all treatment-specific state occupation probabilities across a range of predefined time-points. Our aim is solely to demonstrate that our one-step approach leads to a compatible estimator, respecting the bounds of the parameter space, whereas any approach estimating each univariate component separately is not guaranteed to do so.

We specifically define our target parameter as the vector of treatment-specific absolute risks for both event types evaluated across a grid of follow-up time-points:

$$\Psi(P) = \{\Psi_{1,1}(P), \ldots, \Psi_{1,K}(P), \Psi_{2,1}(P), \ldots, \Psi_{2,K}(P)\} \tag{15}$$

with $\Psi_{j,k}(P) = \mathbb{E}[F_j(t_k \mid A = 1, L)]$ for $j = 1, 2$ and $k = 1, \ldots, K$. Since, for all t, we have $S(t \mid A, L) = 1 - F_1(t \mid A, L) - F_2(t \mid A, L)$, and thus $D^*_{S,t} = -(D^*_{1,t} + D^*_{2,t})$, the one-step targeted estimator that solves the score equations for the two treatment-specific absolute risk functions necessarily solves the score equation for the treatment-specific event-free survival probability too. Table 1 shows the results using the one-step algorithm: the sum of all three estimated state probabilities, $F_1 + F_2 + S$, is ensured to be 1 at all time-points.

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Table 1. *Estimated treatment-specific state occupation probabilities using the one-step algorithm for the follicular cell lymphoma data*

	F_1	F_2	S	Sum
$\frac{1}{4}$ yrs	0.1076	0.0000	0.8924	1.0000
$\frac{1}{2}$ yrs	0.1144	0.0000	0.8856	1.0000
1 yr	0.1216	0.0170	0.8613	1.0000
$1\frac{1}{4}$ yrs	0.1221	0.0170	0.8608	1.0000
$1\frac{1}{2}$ yrs	0.1385	0.0297	0.8318	1.0000
1 yrs	0.1964	0.0356	0.7680	1.0000
$2\frac{1}{2}$ yrs	0.2255	0.0358	0.7387	1.0000

Table 2. *Estimated treatment-specific state occupation probabilities using the iterative algorithm for the follicular cell lymphoma data*

	F_1	F_2	S	Sum
$\frac{1}{4}$ yrs	0.1117	0.0000	0.8883	1.0000
$\frac{1}{2}$ yrs	0.1249	0.0000	0.8751	1.0000
1 yr	0.1337	0.0166	0.8492	0.9995
$1\frac{1}{4}$ yrs	0.1332	0.0166	0.8496	0.9994
$1\frac{1}{2}$ yrs	0.1485	0.0290	0.8212	0.9987
2 yrs	0.2084	0.0346	0.7583	1.0012
$2\frac{1}{2}$ yrs	0.2365	0.0352	0.7292	1.0009

Table 2 shows the results using the iterative targeting procedure of Rytgaard et al. (2023) where each treatment-specific probability is targeted on its own, leading to incompatible estimators not guaranteed to sum up to 1 and to respect the monotonicity constraints.

We further apply the one-step algorithm to estimate the control-specific state occupation probabilities, corresponding to (15) with $\Psi_{j,k}(P) = \mathbb{E}[F_j(t_k \mid A = 0, L)]$ for $k = 1, \ldots, K$. The results are plotted in Fig. 1 along with the corresponding average treatment effect estimates; the figure displays the clear time-varying effect of the treatment that would not have been captured by estimation solely using a univariate causal effect.

7. Concluding remarks

This work provides the methodology for constructing semiparametric efficient substitution estimators for multi-dimensional parameters: as far as we are concerned, there is no other method to achieve this for the considered event history setting. The closest work we are aware of is that of Westling et al. (2020), who proposed semiparametric and efficient estimation of the survival curve by solving the relevant estimating equation for each observed time-point separately and afterwards applying a projection step to yield monotonicity. This procedure neither yields a substitution estimator nor is applicable in the competing risk setting.

Scalability remains so far somewhat of a practical limitation of our methods: for now, we can only target all state occupation probabilities across a limited number of time-points to still solve all efficient influence curve equations. The result of Theorem 1 is important in this regard, telling us that we can get inference for the full survival curve across time by targeting over a grid that is fine enough. Another route, although hampering the substitution property, could be to estimate the subdistributions across a range of time-points and, subsequently, isotonize these estimates as proposed by Westling et al. (2020). On a

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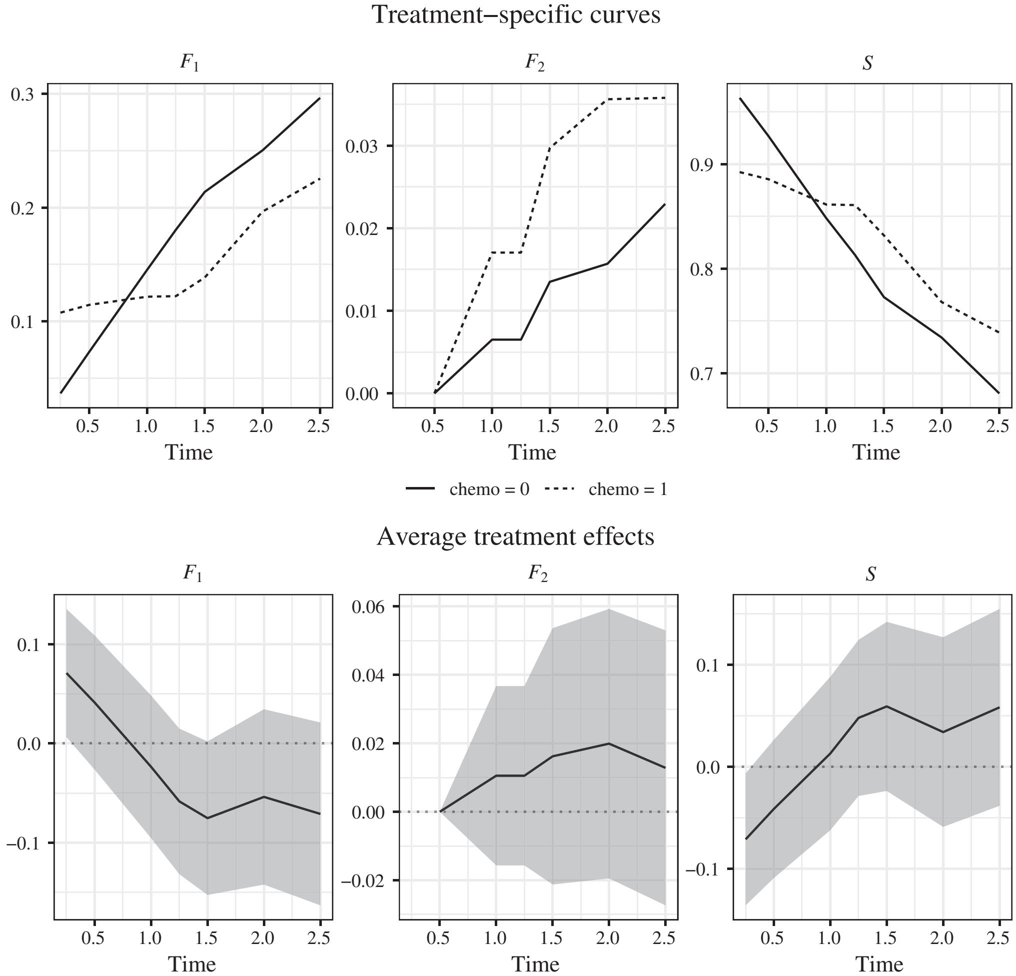

Fig. 1. Estimates from applying the one-step algorithm to the follicular cell lymphoma data. The upper panels show treatment- and control-specific state occupation probabilities, and the lower panels show the corresponding average treatment effects. Time is measured in years.

related note it is an important finding of our empirical investigations that the Σ_n-weighted versions improve computational performance greatly compared to the σ_n-weighted and the unweighted versions, so that this option is absolutely preferred in many situations.

We worked in this paper with event times allowed to take place in $\mathbb{R}_+$, proceeding with the extensions from discrete time as proposed by Rytgaard et al. (2022, 2023). Compared to previously proposed targeting methods, which assume a discrete underlying time grid, our methods allow for analysis of studies with longer follow-up periods where data are collected, e.g., on a daily grid. To move beyond the setting with only baseline treatment and confounding considered in this paper, we suggest coupling our methodology with methods for the more general longitudinal data structure proposed by Rytgaard et al. (2022), to infer the effects of time-varying treatment regimes in the presence of time-dependent confounding.

There are other proposals for causal parameters in competing risk settings that do not involve absolute risk functions. The work by Young et al. (2020) distinguishes between the 'direct effect' and the 'total effect' of a treatment on occurrence of an event of interest, of which our parameters correspond to continuous-time and multivariate versions of the total effect. Direct effects, on the other hand, involve interventions such as to prevent death from other causes and, as such, allows separation of effects on occurrences of competing events. However, inference for the direct effect requires extra strong conditional independence assumptions between latent event times that are most often too unrealistic to provide a practically meaningful interpretation (Andersen & Keiding, 2012). The work by Stensrud et al. (2022) introduces the novel idea of separable effect estimands as an alternative to isolate effects on occurrences of competing events. This work escapes the need for conditional independencies between latent event times by requiring instead that the treatment can be split into disconnected components, each affecting occurrences of exactly one type of event.

Supplementary material

The Supplementary Material includes the proof of Theorem 1, additional comments on the implementation, the derivation of the efficient influence function and additional simulation studies.

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